| □Backend Engineer

010-9101-5429 | digle117@gmail.com | github.com/PreAgile | astro-paper-23v.pages.dev

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NestJS □□ □□ □□□□□ □□□ □□ □, □□□□, □□□ DB □□ □□ □, □□ □□□, □□ □□□ □□ □□□ **Kotlin/Spring/JPA/Kafka/Resilience4j** □□□□ □□□□□ JVM □□□ □□□ □□□□ □□□□. □□·□□□□□□□ □□□ □□ □□□ □□ □ □□□ □□□. **Kotlin □□□ □□□□□ Kotest □ organization □□ □□□ 6□ PR** □ □□□□, **LINE Armeria** □ MDC export □□ PR □, **Spring Batch** □□ 4□ □□ □□□□ □□ chunk scanning □□ □□ PR □ □□□□□□.

□□ □□ □□

□□	Before → After	□□
□□□ □ □□	**□ 800□ → 90□ □ (88.75% ↓)**	□□ IP □□ □□□, □□ □□□
□□ □□□	**70% → 98%**	□□□ □□ □ □□ □□□□
□□ □□□	**99.2%** (□□ 6□□)	□□ □ □□□□□ FIFO + lease TTL
□□ □□ □□	**0□ / 6□□**	Idempotency-Key + □□ □ □□
Akamai □□□ □□□	**77.8% → 100%** (18/18 iter)	Referrer Warming □□ □□
□□ □□□□ webhook race	**□□·□□ □□□ □□**	□□ □□ □□□ □□ □ □□□
□□□□	**Kotest 6 PR □□ (org □□) / Armeria 1 PR □□ □ / Spring Batch · Spring Cloud Gateway □□ □□ + PR □□ □**	GitHub PR □□

□□□□ □□

JVM □□□ OSS □ □□ □□ — □□ **6** · □□ □ **1** · □□ □□ **+ PR** □□ □□.

[Kotest](https://github.com/kotest/kotest) — Kotlin □□□□□ □□□ □□□□□ (4.7k stars)

2026.03 □□□□□ sksamuel □ □□ □□□ **organization** □□ □□, □ □□ 6 PR □□.

#	□□	□□
1	□□ □□ Test Metadata Public API □□ □□ (+355/-16, 5 □□ 16h □ □□)	[issue #5103](https://github.com/kote

		st/kotest/issues/5103), [PR #5905](https://github.com/kotest/kotest/pull/5905)
2	□□□ data class □□ □□ diff	[issue #2545](https://github.com/kotest/kotest/issues/2545), [PR #5835](https://github.com/kotest/kotest/pull/5835)
3	JSON Schema anyOf / oneOf □□ □□	[issue #4463](https://github.com/kotest/kotest/issues/4463), [PR #5807](https://github.com/kotest/kotest/pull/5807)
4	JSON Matchers □□□ Json □□□□ □□	[issue #4601](https://github.com/kotest/kotest/issues/4601), [PR #5795](https://github.com/kotest/kotest/pull/5795)
5	@OnlyInputTypes □□ □□ □□ □□□	[issue #5589](https://github.com/kotest/kotest/issues/5589), [PR #5789](https://github.com/kotest/kotest/pull/5789)
6	shouldHaveSingleElement □□□ □□□	[issue #5755](https://github.com/kotest/kotest/issues/5755), [PR #5756](https://github.com/kotest/kotest/pull/5756)

[Armeria](https://github.com/line/armeria) — LINE □ Java □□ □□□ RPC □□□□□ (5.1k stars)

□□	□□
RequestContextExporter BuiltInProperty □□ + Logback MDC export □□□ □□ (+339/-5, □□□□□ 1/2 □□, □□□ □□ □□ □)	[issue #4403](https://github.com/line/armeria/issues/4403), [PR #6683](https://github.com/line/armeria/pull/6683)

[Spring Batch](https://github.com/spring-projects/spring-batch) — Spring □ □□ □□ □□ □□□□□ (2.9k stars)

□□ □□ □□, PR □□ □.

[illegible]	[illegible]
fault-tolerant step chunk scanning [illegible] [illegible] 4[illegible] [illegible] [illegible] ([illegible] "documentation issue" [illegible] [illegible])	[issue #3946](https://github.com/spring-projects/spring-batch/issues/3946)
conditional flow decider order [illegible] ([illegible] "this is a bug" [illegible])	[issue #4478](https://github.com/spring-projects/spring-batch/issues/4478)

[Spring Cloud Gateway](https://github.com/spring-cloud/spring-cloud-gateway) — Spring [illegible] reactive API gateway

[illegible] [illegible] [illegible], 5.1.x [illegible] [illegible] [illegible].

[illegible]	[illegible]
Apache HttpClient5 [illegible] [illegible] [illegible] [illegible] ([illegible] [illegible] [illegible], [illegible] [illegible])	[issue #3311](https://github.com/spring-cloud/spring-cloud-gateway/issues/3311)
API Versioning Predicates [illegible] [illegible] [illegible] ([illegible] [illegible] [illegible] [illegible])	[issue #4024](https://github.com/spring-cloud/spring-cloud-gateway/issues/4024)

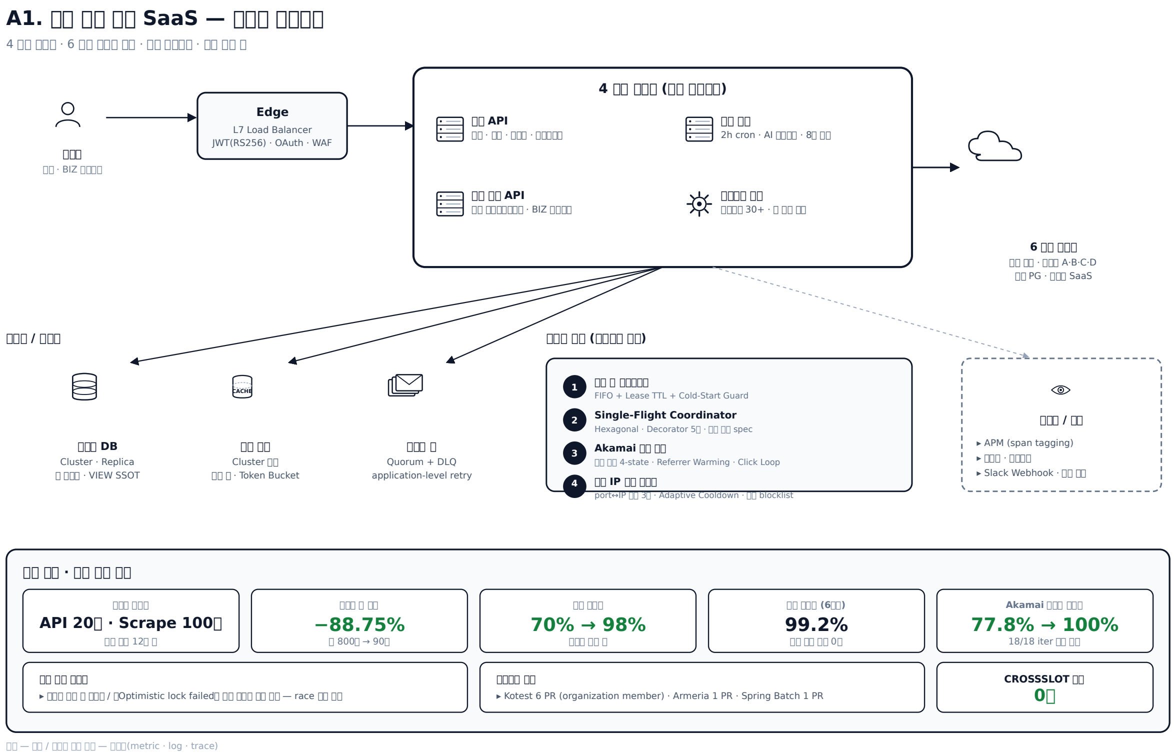
[illegible] — [illegible](Lemong) | | [illegible] 2024.11 ~ [illegible]

[illegible] [illegible]

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] SaaS [illegible] [illegible] [illegible]. [illegible] **API [illegible] 20[illegible] [illegible], [illegible] [illegible] 100[illegible] [illegible],** [illegible] [illegible] **12[illegible]** [illegible] [illegible] [illegible], [illegible] 30+ worker [illegible] [illegible] [illegible]. [illegible] 4 [illegible] [illegible] [illegible].

[illegible]	[illegible] [illegible]	[illegible]
[illegible] **API** [illegible]	[illegible]·[illegible]·[illegible]·[illegible] (BIZ [illegible] ↔ [illegible])	REST API 137 controllers / [illegible] [illegible] 152
[illegible] [illegible] [illegible]	2[illegible] cron [illegible] [illegible] · AI [illegible] · Active-Active 8[illegible] [illegible]	5,700+ [illegible] / 13,000+ [illegible]
[illegible] [illegible] **API**	[illegible] [illegible] [illegible] · [illegible] [illegible] · AI [illegible] [illegible] · BIZ [illegible]	[illegible] [illegible] Hash + ZSet
[illegible] [illegible] [illegible]	6 [illegible] [illegible] [illegible] · [illegible] [illegible] [illegible] · Akamai [illegible] · [illegible] IP [illegible]	[illegible] worker 30+

[illegible] [illegible] [illegible] [illegible] · [illegible] [illegible] (Quorum Queue) · [illegible] [illegible] (Cluster [illegible]) · [illegible] DB (Cluster) [illegible] [illegible], [illegible] APM (span tagging) · [illegible] [illegible] ([illegible] [illegible]) · Slack Webhook ([illegible] [illegible] · cron [illegible]/[illegible]/[illegible]) [illegible] [illegible] [illegible] [illegible].



A1. □□ □□□ □□□□

□□□ □□·□□(A) → □□ □□ □□□·□□ □□□(B) → □□ □□□ □□·□ □□ □□(C) → AI □□□□□·□□ □□□(D) □ □□□ 15 □ □□□ □□ → □□ → □□ □□□ □□□□□.

□□□ A. □□·□□

□□ 1. □□ webhook 4□ □□□ — □□ □ (ABA-safe) · DB UNIQUE · □□□□ · | □ □□ 2025.09 ~ □□ □

□□ — □□ PG webhook □ □□ □□ □□□□ □□ □□□□ □□ 5□.

- PG □□ retry (5xx / timeout)
- □□□□ schedule 1□ □□ □ fallback retry → PAID / FAILED □□ □□ □□
- □□ □□ □ □□□
- □□ □□ □□ □□
- Load Balancer keep-alive □□□

□□ INSERT □□ □□ row □□ □□ → □□ □□ → □□□ 2□ □□ → □□□□ □□ □□ → □□ □ □□ □□□□ □□□.

□□ □□□□□ □□□□□ in-memory dedupe □□□. ORM findOne → save □□□ read-modify-write □□□ race window □□. □□ □□ □ □□□□□ TTL □□ □ ABA □□(Martin Kleppmann Redlock □□) 가 □. DB UNIQUE □□□□□ PG retry □□□ 1□ □□ □□. □□□□□□□□ □□ INSERT race □□ □□.

□□ — □□ □□□□ □□ race □ □□ □ □□ **4**□ □□□ □ □□□□ □□□□□□.

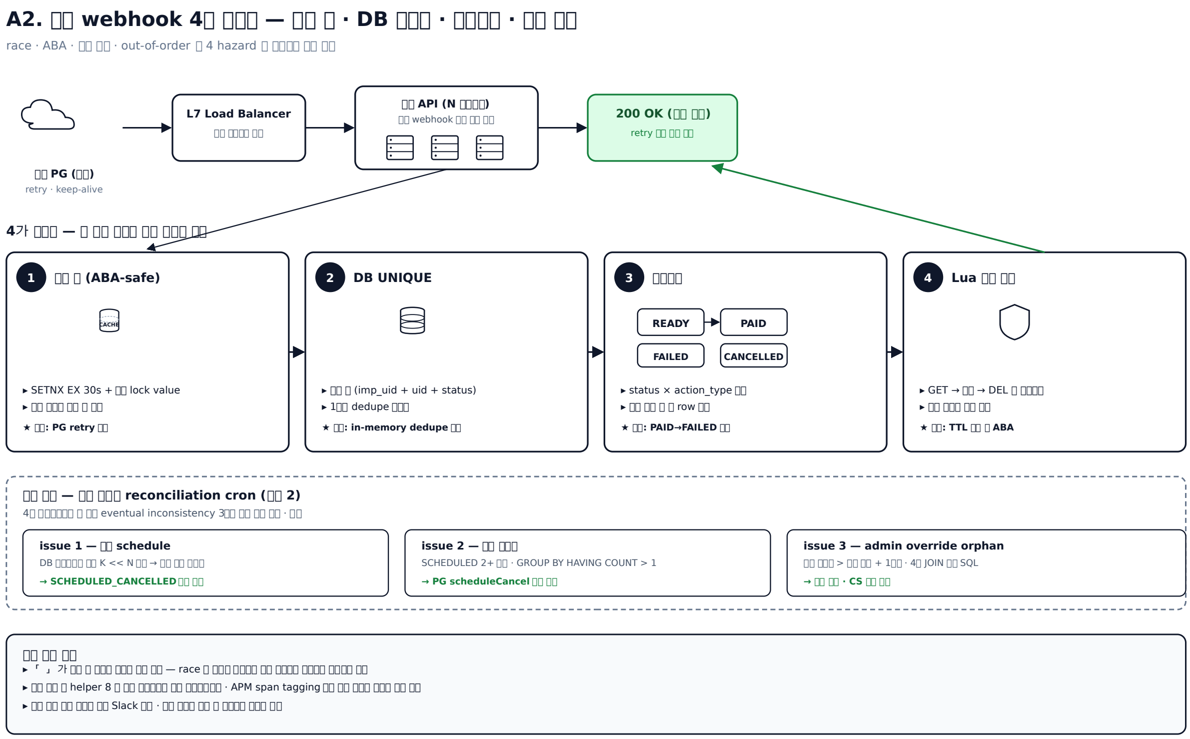
1. □□ □ **(ABA-safe)** — □□ □□ SETNX EX 30s + □□ lock value · Stripe Idempotency Key □□□□ □ □□□ lock value □ ownership □□

2. **DB UNIQUE** — □□ □ imp_uid + merchant_uid + status · 1□□ dedupe window · □□ □□ fail-open □□□ □□ □□

3. □□□□ — READY → PAID / FAILED / CANCELLED × action_type □□ · □□ □□(PAID □ FAILED □ □) □ □□ row □ □□

4. **Lua □□ □□** — GET → □□ → DEL □ □□□□ · □□ □□□ □□ □□, TTL □□ □ ABA □□

□□□ SETNX + Lua DEL helper □ □□ □ 8 □ □□ □□(promotional upgrade · change plan · verify code · □□ □□)□□ □□ helper □ □□□□□□.



A2. webhook 4□ □□ □□□

□□

- 「가 」 □ □□□ □□□□ □□□□□ □□□□ race 가 □□□ □□□□ □□ □□□□ □□□□ □□□□ □□.
- □□ □□ □□ □□□ □ Slack □□ □□ — □□ □□□ □□□□ □□□.
- APM span tagging □□ □□ □□□ □□□(amount_mismatch, coupon_select)□ □□ □□.
- □□ □ helper(delIfValueMatches)가 □□ □ 8 □ □□ □□(promotional upgrade · change plan · verify code · □□ □□ □)□□ □□ helper □ □□□.

JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

□□ 4□ □□□□ Java 21 / Spring Boot / JPA / Redisson □□ □□□□□□ **ADR-006** □□ □□□ **4□□** □ □□□□□□□.

- ① □□ □ — Redisson watchdog ([ADR-004](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/adr/ADR-004-distributed-lock.md)) — Redisson vs SET NX+Lua □□ □ □□
- ② DB UNIQUE — JPA @Version + OptimisticLockException + MySQL ER_DUP_ENTRY → @RestControllerAdvice □□
- ③ □□□□ — idempotency_key □□□ + 4 □□(RECEIVED→PROCESSING→SUCCESS/FAILED)
- ④ □□□□ □□ — @Transactional(REQUIRES_NEW) + □□ □□ OSIV=false ([ADR-BE-008](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/adr/ADR-BE-008-transaction-boundary.md), [EXP-09b 9 □□□□ □ □□ □□](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/experiments/EXP-09b-tx-boundary.md))

EXP-02 □□□□ □□ □ 4□ (□□□ · □□□ · GET_LOCK · Redisson) □□ □□ — GET_LOCK □□ □□.

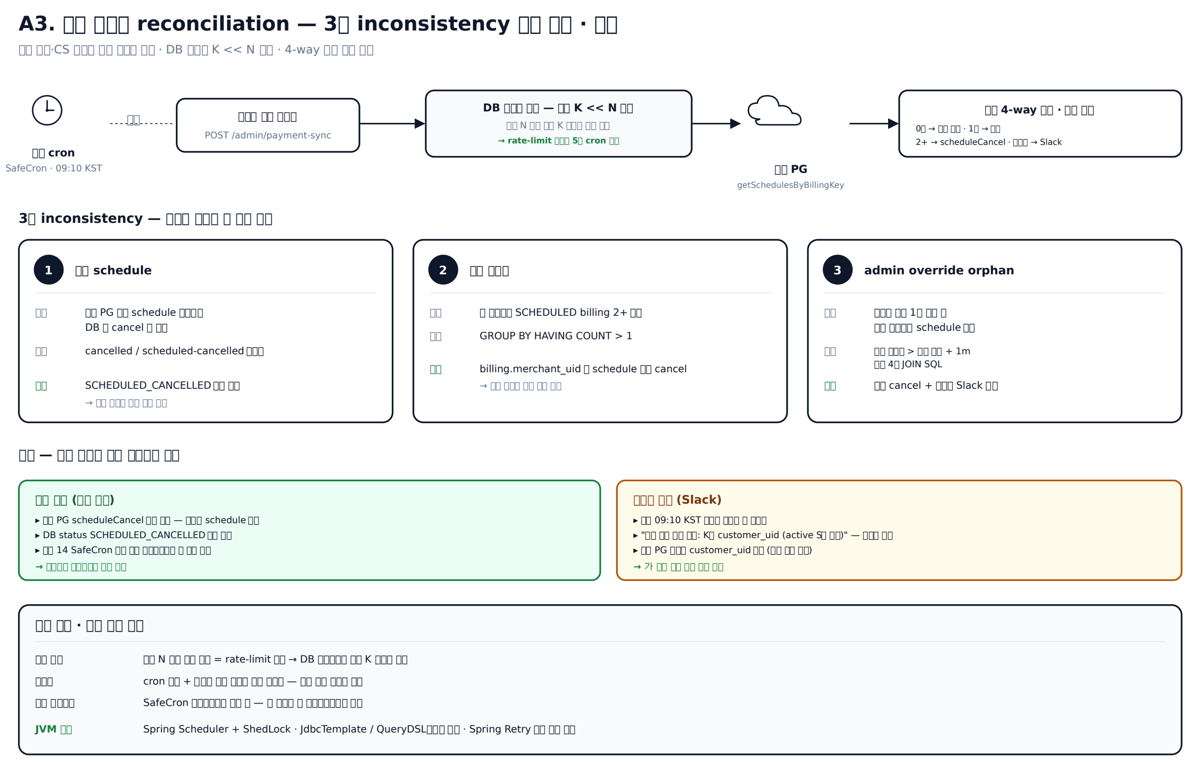
□□ 2. □□ □□□ reconciliation — □□ schedule · □□ □□□ · admin override orphan | □ 2025.10 ~ □□ □

□□ — □□ 1 □ 4□ □□□□□□□ □ □□ 3□□ silent □□□ □□□ □□□□□.

- **□□ schedule** — □□ PG □□ schedule □ □□□□□ DB □ cancel □ □□ → □□ □□□ □□ □□
- □□ □□□ — □ □□□□ SCHEDULED billing 2 □ □□ □□ □□ □□□ □□ □□
- **admin override orphan** — □□□□ □□□ 1□ □□□ □□ □□□□□ □□ □□□□ schedule □□ → active □□ □ □ □□

□ □ □□□ □□□ / □□ □□□ □□□ □□□ □□□□ □□□□□. □□ PG API □ □□ □□□ □□ □□□□ □□ □□□ □□ □□□ □□□□□, DB SCHEDULED □ PG schedule_status □ □□□ eventual consistency □□□□□.

□□ — **□□□ □□ K << N □□ □□ → □□ □□ □□□ → 4-way □□ □□ □□**」 □□□ □□ cron + □□□ □□ □□□ □□ □□□ □□ □□. DB □□□□□ □□ customer □ □□□, □□□ K □□ □□□□ □□ PG schedule API □□, □□□ 4-way □□ □□ — schedule 0□□ DB □□□ □□, 1□□ □□, 2□ □□□□ □□ schedule □□ cancel, API □□□□□ Slack □□. admin override orphan 「 □ □□□ □□□□ □□ □□□□ □□□□ 1□□ □□ □□ □□□ □□ SQL □ □□□□□. □□ cron □ SafeCron □□□□□(□□ 14)□ □□ □□□□□□□ □ □□ □□□□□ □□ □ □□.



A3. □□ □□□ reconciliation — 3□ inconsistency □□ □□

□□

- 3□ inconsistency(zombie_schedule / duplicate_billing_key / admin_override_orphan) □□ □□ □ □□ □□.
- □□ □□·**CS** □□□ □□ □□□ □□ — □□ 09:10 KST □ □□□ □□□ Slack □ □□□□ □□.
- 「 □ □□□□□ □□ □□ K□ (active S□ □□)□ □□□ □□□ □□ — □□ PG rate-limit □□□ 5□ cron □□ □□.

- □□ SQL □□□ admin override □□□ □ □□ □□.

JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be) §5.5)

□□ reconciliation □ ADR-006 □ 4□ □□□ □□□ layer □ □□□□.

- Spring @Scheduled + **ShedLock** (Redis/JDBC backend) □□ □□□□ cron □□□
- QueryDSL □ DB □□□ (GROUP BY HAVING COUNT > 1, DATE_ADD INTERVAL □□)
- Spring Retry + Resilience4j RateLimiter □ □□ PG getSchedulesByBillingKey □□
- ApplicationEventPublisher + @TransactionalEventListener □ Slack □□ □□□□ □ □□

□□ 3. □□ □□ □□ — □□ □□ · □□□□ · schedule fallback □□□ 3□ idempotency □ 2025.11 ~ □□ □

□□ — □□□ □□ □□□ □□ □□ □□□□. □□□□ schedule □□ □□□ □□□ amount □ □□ □□ □□□ □□ □□□ □□□□ silent failure 가 가 □□□□□. schedule □□ □ □□ A □ 50% □□ amount □ □□□□□ □□ □ □□ □□□ □□ A 가 □□/□□□ □ □□ PG □ □□□ amount 가 → □□□□ □□. 가 가 □□□ □□ API 가 가 □ □ □□□□ □ □ □□ □□ □□ □□□ □□□ □□. □□ □□□□□ + 가 □plan + □□□ user □□□ □ □□ □□□□ □□□ API □ □□ update □ □□ □ □□□ □□□□ □□□ □.

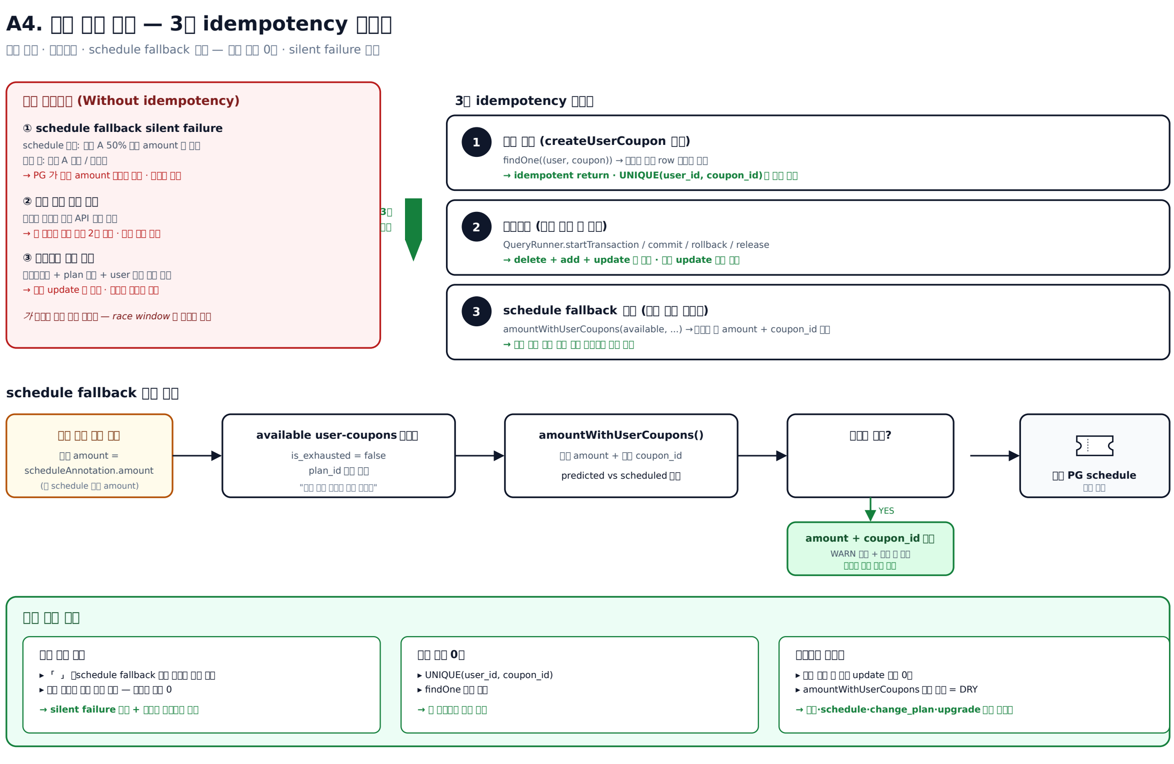
□□ — **3□ idempotency □□□**:

1. □□ □□ — createUserCoupon □□ □ findOne((user, coupon)) □□ □□ row □□, □□□ □□□ □□ (idempotent return)

2. □□□□ — ORM QueryRunner □□ □□□ □□□□(startTransaction / commit / rollback / release) — delete + add + update □ □ □□□ □□

3. **schedule fallback** □□ — schedule 가 □□ □□□□user-coupons 「 □□ □□□ □□ □□□ □□□ □□□ → □□□ □ amount + coupon_id □□

amountWithUserCoupons □□ □□□ □□·schedule·change_plan·promotional upgrade □□ □□□□ □□□ □□ DRY □□.



A4. □□ □□ 3□ idempotency □□□

□□

- □□ □□□□ □schedule fallback 」가 □□ □□□ □□□ □□□ □□ □□ □□ □□□.

- □□ □□ □□ □□ 0□ — UNIQUE(user_id, coupon_id) + findOne □□ □□□ □□ □□.
- □□ □□ □ □□ update □□ 0□.

JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

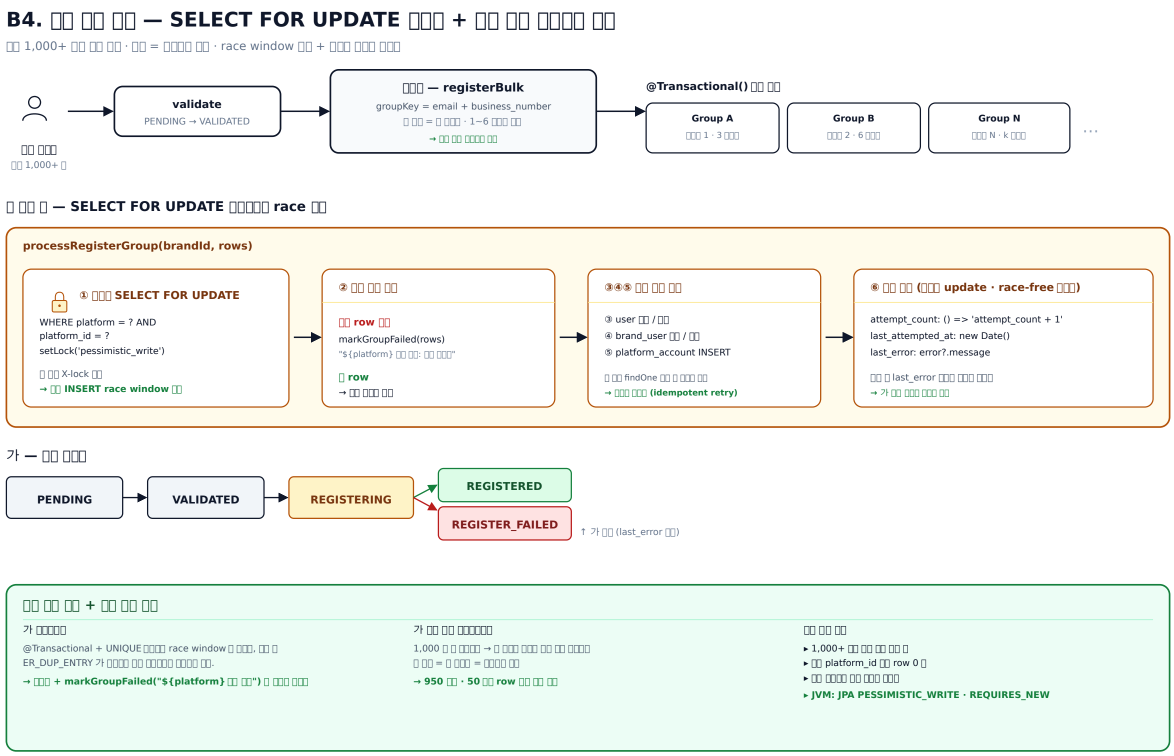
- □□ □□ — JPA findByUserAndCoupon idempotent return + DB UNIQUE(user_id, coupon_id) □□ □□
- □□□□ — Spring TransactionTemplate □□□ □□□□ + JPA @Version □□□ ([ADR-BE-007 □□□□](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/adr/ADR-BE-007-isolation-level.md), [EXP-03 @Version □□□ silent Lost Update 74□ □□](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/experiments/EXP-03-optimistic-lock.md))
- schedule fallback □□ — @PostLoad + @PrePersist Hibernate □□ + Bean Validation @AssertTrue

[illegible] B. [illegible] · [illegible]

□□ 4. □□ □□ □□ — □□□ X-lock + | □□□□ □□ □ 2025.12 ~ □□ □

□□ — □□□ □□~□□ □ □□□ □□ **1□ = 1 □□□ = 1~6** □□□ □□ □□□□ □ □□ □□□□□. □ □□ □□ INSERT □ 1,000□□ □□ □□ □□□, □□ (platform, platform_id) 가 □ □□ □□□(□□□ □□) 가 □□□□ □ findOne → save race □ □□ row 가 □□□□. ORM □□ □□□□ REPEATABLE READ + UNIQUE □□□ □□□ race window 가 □□□□ ER_DUP_ENTRY □□ □□□□ □□□ □□□□ □□□□, 1,000□□ □ □□□□□□ □□□ □ □□□ □□□ □□ □□□□ □□ □□/가 row □□□ □□□□ □□□□.

□□ — **`SELECT ... FOR UPDATE` □□□ + □□ □□ □□□□ □□ □ □□□ □□.** □ □□□(groupKey = email + business_number) □□□ □□ @Transactional() □□□□ □□□□ □□ = □□□□ □□. □□□□ □□□ (platform, platform_id) □□ X-lock □ □□, □□ □ □□ □□□ □□□ □□□ □□□□ marking. □□□ □□□□ □□□ findOne □□ + □□ □ update(attempt_count + 1)□ race-free □□□. □□□□□ PENDING → VALIDATED → REGISTERING → REGISTERED | REGISTER_FAILED 가 □ □□□ last_error · last_attempted_at · attempt_count □ row 가 □□□□ □□□□ □□ □□□ □□□□ □ □□ □□□□.



B4. □□ □□ □□ — □□□ X-lock + □□ □□ □□□□ □□

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- 1,000+ □□ □□ □□ □□, □□ □□/가 row □□□ □□□ □□.
- □□ platform_id □□ □□ □ □□ row **0**□.
- 「 1,000□ □ 950 □□, 50 □□□ □ □□□ 50□□ □□□ □□ — □□ □□ □□ □□.

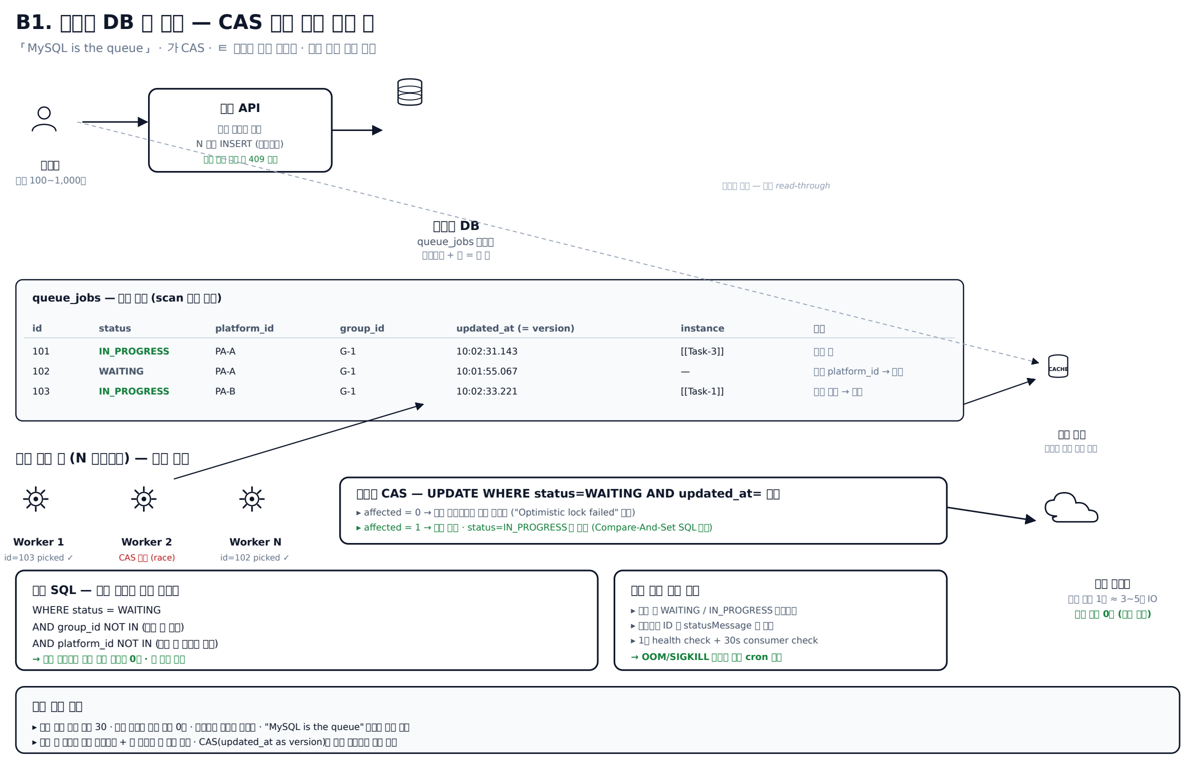
JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

- JPA LockModeType.PESSIMISTIC_WRITE (SELECT ... FOR UPDATE) + Spring @Transactional(propagation=REQUIRES_NEW) □ □□ = □□□□ □□
- EXP-02 □ 4□ □□ — □□□ / □□□ / GET_LOCK / Redisson □□ □□ □ □□□ □□ □□ □□□
- EXP-14 IDENTITY □□ □□□ saveAll batch insert □□□ □□ — □□ □ TABLE/SEQUENCE 가 □□

□□ 5. □□ □□ □□ — □□□ DB □ □□ □□ CAS | □□ □ □ 2025.10 ~ □□ □

□□ — □□□□ □□□ □□□ □□ 100~1,000□□ □ □□ □□ □□ □□□ □ □□ □□□□□. □□ □□□ □□ □ □ □□□ **3~5□ IO** — HTTP □□ □□□ Load Balancer 5□ □□□□ □□, □□ □□□□ in-memory queue □ □□□ □ □□ □□, 가ᄐ가 ᄐ□ □□ □□□□ worker 2□□ □□ □□□□□ □□ □□□ □ □□□ □□ □ □□ □□□ □□□□□. □□ □□□ □□ stateful □□·□□·□□·□□□ □□□ □ □□(fire-and-forget □ □□), □□ □ □□□□□ □□□□□ □□ □□ □□□□□.

□□ — **□□□ DB** □□□□ □□ □□ □ □□□□□ □ □□□ □ □□ □□, □□ □□ □□ □□□ **ORM □□□ = CAS(Compare-And-Set) □ SQL** □□ □□ □□. updated_at □ version □□□□ □□□ UPDATE WHERE id=? AND status=WAITING AND updated_at=□□ □ affected=0 가 가 가 ㅇ □□□□□□□. □□ □□□ □□ □ □ □□□ □□ SQL □ platform_id NOT IN (□□ □ □□) □□□□ □□ — □ □□□ □□ = □□□ □□□ □□. □□ □□ □□ □□□ □□ □ WAITING/IN_PROGRESS □□ □□□□ + □□□□ ID □ statusMessage □ □□ □□□□ □□. □□□□ □□ □□ + acquireProcessingLock □□ □□□ □□ □□ □□.



B1. RDS Queue + □□□ □□

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- 「 □ Job N modified by another instance (optimistic lock failed)□ □□ □□ — race □□□□ □□□□ □□□□ □□.
- □□ □□□ □□ □□ **0**□, □□□□ □□□ □□□.
- □□ □ □□□ □□ □□ □ □□□ □□ □□□□□ □ □□□ □ □□ □□ □□ □□□□ □□□.

JVM/Spring □□□ ([commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator))

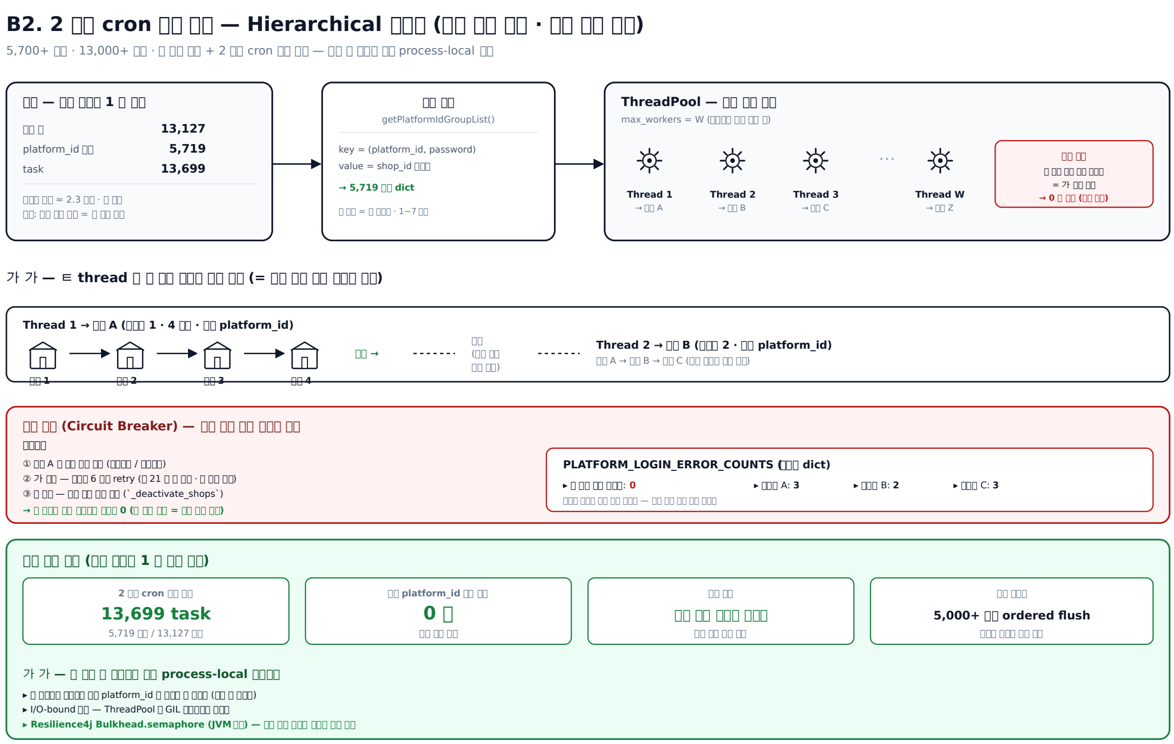
RDS polling □ □□□ Kafka outbox □ □□□.

- [ADR-005 Outbox □□ → Debezium CDC](https://github.com/PreAgile/commerce-batch-orchestrator/blob/main/docs/adr/ADR-005-outbox-cdc.md) — Dual-Write □□ □□
- [ADR-MQ-008 Consumer □□ + DLQ + retry topic](https://github.com/PreAgile/commerce-batch-orchestrator/blob/main/docs/adr/ADR-MQ-008-consumer-idempotent.md) — At-least-once + □□ = Effectively-Once
- JPA @Version □□□ + OptimisticLockException = CAS SQL □□, Micrometer gauge □ □□□ □□□

□□ 6. 2□□ cron □□ □□ — 5,700+ □□ × 13,000+ □□ hierarchical □□□ □ 2025.10 ~ □□ □

□□ — □□ □□ □□□□□ 6□ □□ □□□□□ □□ □□□ □□□ 2□□ □□□ □□□ □□□□□. □□ □□ □□ □□ □□□ 1□ □□□□ **13,127 □□ / 5,719 platform_id □□ / 13,699 task** □□ □□□□□. □□ □□□□ □□ □□□□ □□ □□ □□ □□ □ captcha / □□□□ / □□ □□□□□, □□ □□ ThreadPool □ □□ □□ □□□ □□ □□ □□ □□□ = □□ □□, □□ □□□ 13,000 × 3□ = 11□□ → 2□□ cron □□ □ □□□.

□□ — 「 □□ □□ / 」 □□□ □□**hierarchical** □□□ □□. □□ □ □□□ □□ process-local □□□□ □□□□□ □□ — (platform_id, platform_password) □□ □□□ □□ □□□ ThreadPool □ □□□□ □ □□□ □ worker thread 가 □□□□ □□□ □□ □□□□□. I/O-bound(HTTP/DB)□ thread 가 □□. □□ □□ □□ □□□□□ □□□□ □□□□ □ □□□ □□□ □ □ □□. □ □□ □□ □□ □ □□ □□ □□ □□ — □ □□ □□□ □□□ □□□ 100% □□ □□□□ □ □□ □□ □□□ □□ □□ □□(circuit breaker) □□.



B2. Hierarchical □□□ (□□ □□ □□ · □□ □□ □□)

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- □□ platform_id □□ □□ **0**□ (□□ □□ □□).
- □□ □□ □□ □□□ □□□ □□, □□ □□ □□ □□ — □□ □□ □□.
- 5,000+ □□ □□ □□□□□ ordered logBuffer flush 가□□ □□□ □ □□ □□ □ □□.□□□

JVM/Spring □□□ ([commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator) +

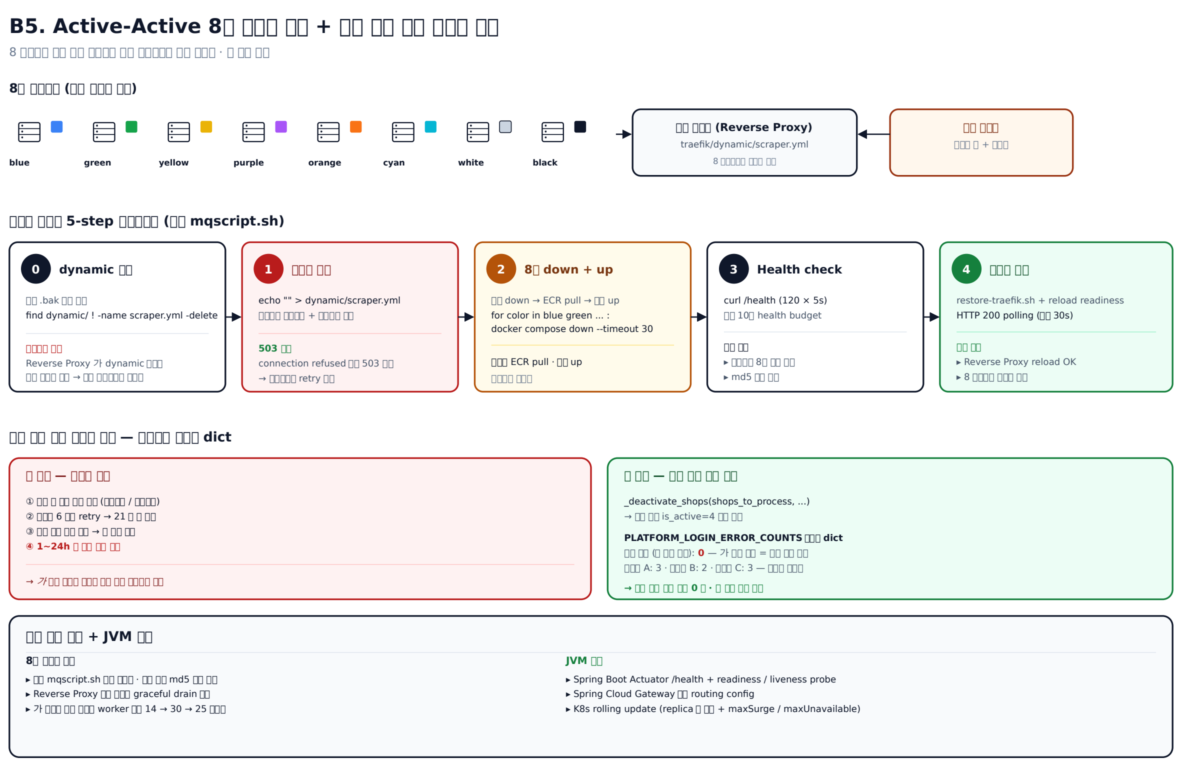
[commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

- Spring Batch + **ShedLock** □□ □□ □□□□ cron □□ □□
- [ADR-MQ-007 Reader 4□ □□](https://github.com/PreAgile/commerce-batch-orchestrator/blob/main/docs/adr/ADR-MQ-007-reader-4.md) → QuerydslZeroOffsetItemReader □□ (E-MQ-04~07 100□ row □□ □□)
- [ADR-MQ-009 CooperativeStickyAssignor (KIP-429)](https://github.com/PreAgile/commerce-batch-orchestrator/blob/main/docs/adr/ADR-MQ-009-rebalance.md) — Rebalance stop-the-world □□
- Resilience4j Bulkhead.semaphore □ □□ □□ □□□ □□ + CircuitBreaker □ □□□ □□

□□ 7. Active-Active 8□ □□□ □□ + | □□ □□□ □□ □ 2026.02

□□ — □□□□ □□ □□□□□ □□ □□□ □□□□ □□□ 8□(blue/green/yellow/purple/orange/cyan/white/black) Active-Active □□□ □□. 8□ □□□□ □□ down/up □□ traffic gap □□ 503 □ □□ □□□□□. □□ □□ □□□□ □□□□·□ □□□□ □□□ throw 가 □, □□ □□ □□ □□□ □□ □□□ □□ □□□□ □□□ □□(가가 ㅌ □□ → □ □□ □□ → □□□□□ □□□ □□ 7□□ □□ □□ → □□ retry □ 21□ □ □□)□ □□ □ □□□□□.

□□ — □□ □□□ □□ □□ □□ graceful drain □ **5-step** □□□ □□□□□ + □□□□ □□□ **dict** □□ □□ □□ □□ □□. dynamic □□ □□ → □□□ □□ □□□(□□□ □□, □□□□□ □□□□ → 503 □□) → 8□ □□ down + □□ up → health check(120 × 5s = 10min budget) → □□□ □□ + reload readiness □□. 가 □□□ □□□□ □□□ 0(□ □□ □□ = 가 □ □□)가 □ □□ □ □□□□□□□□□.



B5. Active-Active 8□ □□□ □□ + □□ □□ □□ □□□ □□

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- 8□ □□□□ □□ □□ □□□□ □□ □□□□□ □□ □□□ □□□.
- □ □□ □□ — □ □□ □□ □ □□ □□ □□, 가ㅌ 가 □□ **0**□.
- 가 □□□ □□□□ worker □ □□ 14 → 30 → 25 □□□.

JVM/Spring □□□ ([commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

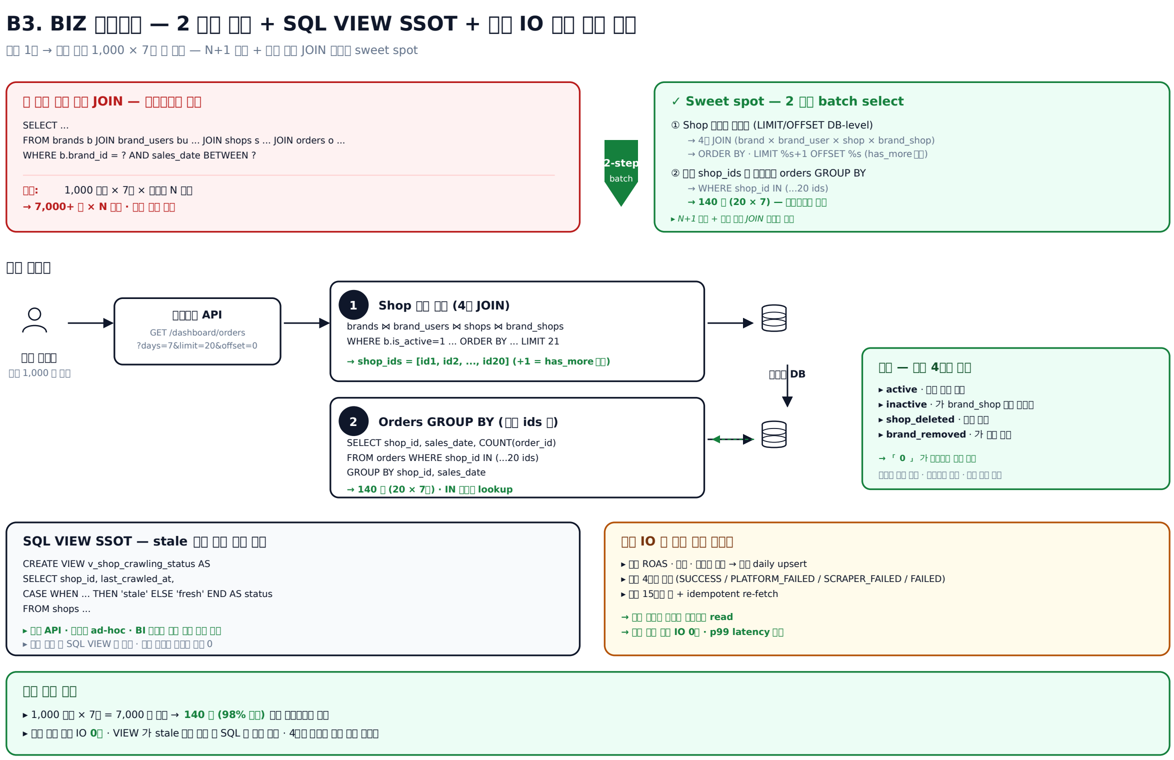
- Spring Boot Actuator /health + readinessProbe / livenessProbe + Micrometer
- Spring Cloud Gateway dynamic routing config + K8s rolling update (maxSurge / maxUnavailable)

- Resilience4j CircuitBreaker □□□ dict □□ — [ADR-007 50%/70%/90% □□ □ 70% □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-007-cb-threshold.md) (CB-1/2/3 □□)

□□ 8. BIZ □□□□ 1,000□□ × 7□ — 2□□ □□ + DB-level pagination + SQL VIEW SSOT □ 2026.01

□□ — BIZ □□□ □□ 1□□ □□ 1,000□□□ □ □□□□ □□□. □□□ □□ 7□ □□ □□, □□□ □□ ROAS / □□ / □□, □□□ □ □□ 7□, □□ □□□ □□(stale) □□, □□□ □□□□ □ □□□ □□□ □□□. □□ JOIN □ □□□ brand → … → orders □□ □□ □□□□□ □□(1,000 × 7 × □□□ N) + □□ □□/□□ IO □ □□ □□□ □□ □□ □□□ □□□□□. ORM □ □□□□□□ 4□ □□ + dynamic where + dynamic order by + dynamic LIMIT □ □□□□□ □□□□ □□□□□.

□□ — 「**select shape □ N+1 □ □□ □□ JOIN □□□ sweet spot □ 2-□□ batch select □ □□□**」 + **SQL VIEW □ □□ SSOT □ + □□ IO □** □□ □□ □□□. 1□□□ shop □□□ □□□(brand × brand_user × shop × brand_shop 4□ □□, LIMIT +1 has_more □□□□ DB-level pagination), 2□□□ □□□ ~20 shop_ids □ □□□□ orders GROUP BY □ □□□□□□ □□(140□ = 20 × 7). □□□ **4□□**(active / inactive / shop_deleted / brand_removed)□ □□□ □□ □□□□ □□□ 0□ □□□□ □□ □□. stale □□ □□□ SQL VIEW □ □□ □□ API · □□□ ad-hoc · BI □ □□ □□□ □□(single source of truth). □□ □□/□□□ □□ daily upsert + 4□□ □□□□ □□□□ □□ □□□ □□□ □□□□ read.



B3. BIZ □□□□ 2 □□ □□ + SQL VIEW SSOT

□□

- □□ 1,000 × 7□ □□ □□ □□□ □□□□ 2□□ □□ + DB-level LIMIT/OFFSET □□ □□□□□ □□ — □□ □□ □□ IO **0**□.
- VIEW □□ □□ □□□ stale □□ □□ □ SQL □ □□.
- □□ 15□□ □ + □□ idempotent re-fetch □ □□ □□□ □□.
- 「 raw SQL □ □□□ □ □□ □□□ □□ □□□□ □□□.

JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be))

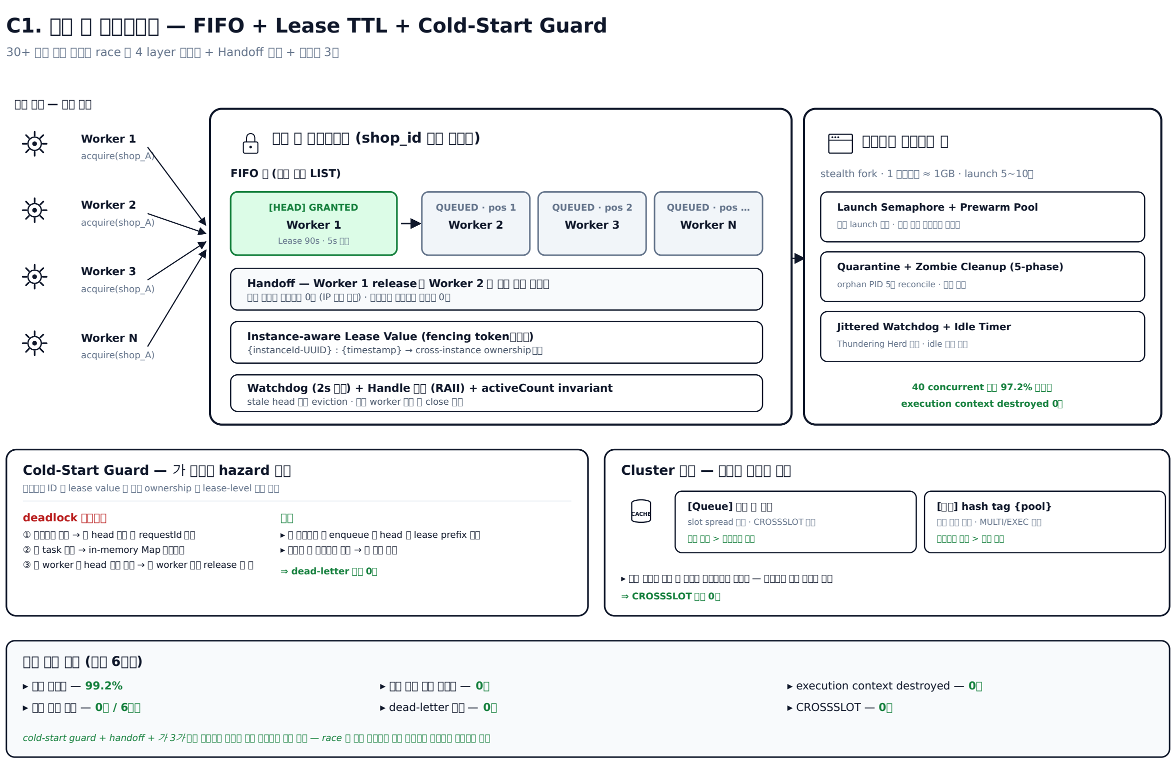
- Spring Data JPA + **QueryDSL** □□ □□ / dynamic order by ([ADR-BE-009 No-Offset Cursor Pagination](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/adr/ADR-BE-009-no-offset.md))
- [EXP-07 1M row OFFSET 171ms → Cursor □□□ □□](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/experiments/EXP-07-pagination.md) — row constructor □□ □□
- @Subselect □ SQL VIEW □□ + @EntityGraph □ N+1 □□ ([EXP-04 N+1 4-depth 121 prepStmt → JOIN FETCH 1](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/experiments/EXP-04-n-plus-one.md))

[illegible] C. [illegible] · [illegible]

□□ 9. □□ □ □□□□□ — FIFO □ + lease TTL + cold-start guard □ 30+ worker □□ □□□ race | 2025.11 ~ □□ □

□□ — □□ □□ 30+ □□ □□□ □ consumer □ □□□□□. □ □□(shop_id) □ □□ API(getReviews, addReply, getStores, keepAlive)가 가 □ □□□□□□ □□□□ □□□ □□ worker 가 □□□ □□□ □□□. □□ □□ □ page □□ □□ navigation □ execution context destroyed, 가는 □□ □□ □□□ □ □□ □□ □□□ IP □□ □□ □□(1~24□□ □ □□ □□ □□ □□). □□□ □□□□ □□ mutex □□ □□□□ □□□□ — □□□□ □□□□ 1 □□□□ ≈ 1GB+, launch 5~10□ → race □ N□ □□□·□□□ □□. □□□□ □□□□ in-process Map □□□ □□□□, □□ □□ Cluster □□□ □□ □□□□□ □□ □ □ □□ CROSSSLOT □□□ □□□□. 가 □□□□ hazard □ **cold-start deadlock** — 가 □□□ □□□□ in-memory Map □ □□□□□ □□ □□ □ head □ □ □□□□□ requestId □, □□□ worker 가 □ head □□□ □□□□□ □ worker □ □□□ □□□ release □□ □□□□.

□□ — **4 layer** □□□: ① FIFO □(□□ □□ LIST) ② per-request lease(90s TTL, 5s □□) ③ **instance-aware lease value** = {instanceId-UUID} : {timestamp} (Martin Kleppmann □ fencing token □□□ □□□ — □□□□ □□□□ lease value □ □□ cross-instance ownership □ lease-level □□ □□) ④ handoff pattern (release □ □□ □ entry 가 가는 □□ □□□). Cluster CROSSSLOT □ □□□ □□□ □□ □□ □ □□ — queue □ □□ □ □□□□ □□(□□ □□ □□), reputation(□□ 10)□ hash tag {pool} □ □□ □□ □□(□□□□ □□ □□). □□ □□□ □□ □ □□□ □□□□□ □□□ □□ □□□ □□□□ □□ □□□ □□□ □□. attach/release Handle □□(RAII)□□ activeCount invariant □ □□□□, pendingClose 5□ □□(immediate / defer / idle-timer / hasWaitingRequests handoff / release)□□ □□ worker □ □□□ □□□ □ close □ □□, □□ release □□□ close. □□□□□ forceRelease(□□ kill switch), forceTerminate(activeCount □□ 0), evictStaleHead(2□ □□ head □ lease □ □□ □ LREM) 3□.



□□

- □□ □□□ **99.2%** (□□ 6□□).
- □□ □□ □□ **0□ / 6□**□ — □□ □ + Idempotency-Key □□.
- execution context destroyed □□ **0**□.
- □□□□ □□□ □ dead-letter □ □□ **0**□ — cold-start guard □□ □□.
- CROSSSLOT □□ **0**□ — hash tag / □□ □ □□ □□□ □□ □□ □□.
- □□ □□ □□ □□□ **0**□.

JVM/Spring □□□ ([commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

- [ADR-SCR-009 3-Layer □□ □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-SCR-009-session-cache.md) — Memory L1 → Redis L2 → □□ □□ L3
- Kotlin Coroutines Mutex + withLock per-key □ in-process FIFO layer
- Redisson RFairLock (FIFO + lease auto-extend) + @PreDestroy graceful release
- Micrometer gauge □ □ □□ / □□ hit rate □□ (SES-1 3□□ hit rate □□)

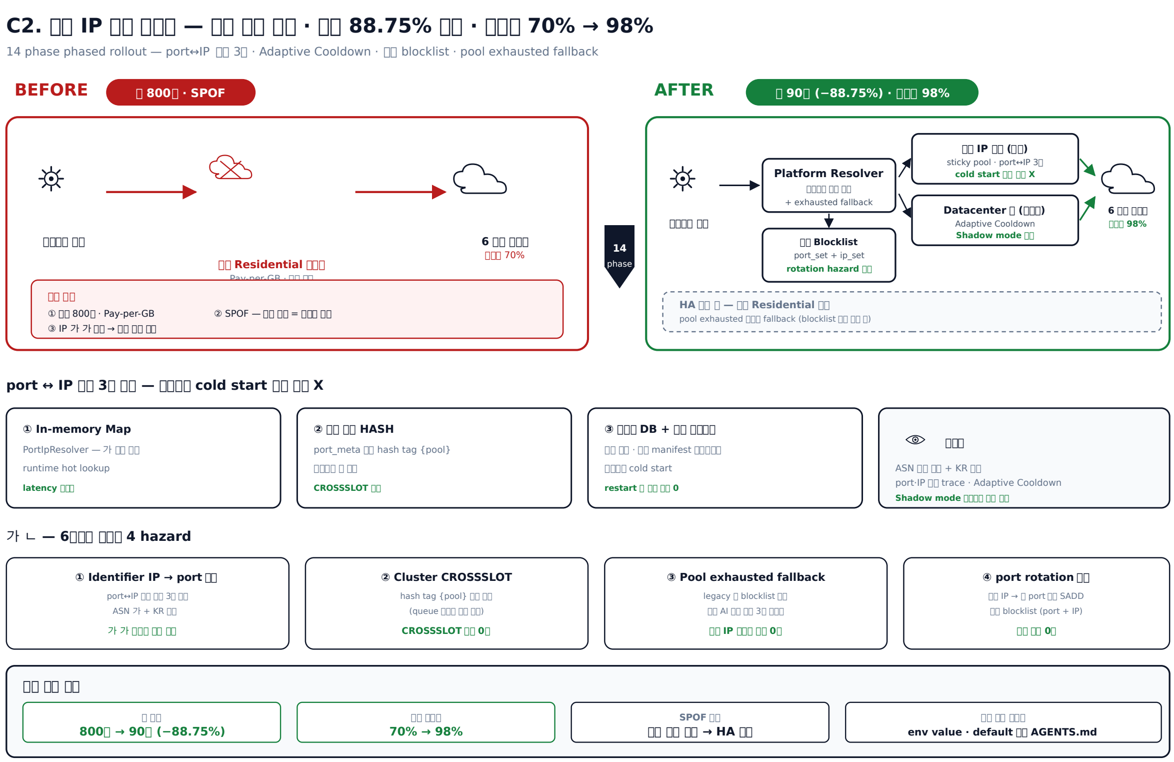
□□ 10. □□ IP □□ □□□ — □□ □□ □□ + □□ 88.75% □□ (□ 800□ → 90□) □ 2025.07 ~ □□ □

□□ — □□ Residential □□□ □□ □□□□ □□ / SPOF / □□ □□□ □ □□ hazard 가 □□□□ □□□□□.

- **□□ / SPOF** — □□ 800 □ □ · □□ □□ □□ = □□□ □□
- **Identifier □□□** — Datacenter □□□□ port □□ IP 가 가ㄴ □□ □□ → □□ □□ □□(IP)□ stable identifier(port)가 □□□
- **Cluster CROSSSLOT** — □□ □□□ · STREAM · blocklist □ □ MULTI/EXEC □ □□ □ □□ □□
- **port rotation □□** — □□ IP 가 □□□ □□ port □ mapping □□ □ port 가 □□ blocklist □ □□

6□□□ □□ 14 phase phased rollout □□ □□□□□.

□□ — **14 phase phased rollout**. □ □□ □ □□□ □□□ □ phase □□ → □□ → □□ □□ → □□ phase □ □□□ cutover. □□ □□ 4가: ① **port↔IP □□ 3□** □□ (In-memory Map + □□ □□ HASH + □□□ DB + □□ □□□□ □□□□□ manifest) — □□□□ cold start □ □□ □□ □□ □□ ② □□ **blocklist** — port_set(□□ □□ port □□ □□) + ip_set(□□ □□ IP □□ □□), allocator 가 port □□ □ lookup □□ IP □□ → ip_set hit □ □ port □ □□ port_set 가 → **port rotation □□ hazard** □□ ③ **Adaptive Cooldown** — 5□ outcome(success / block / timeout / networkError / authError / siteChange / unknownError) × consecutiveFailures × latency 가 health □□ → ranked □□, Shadow mode □ □ □□ □□ □□ □□ □□ ④ **Pool exhausted → legacy fallback** — □ □□ exhausted □ legacy path □ □□, legacy □ blocklist □□□ □□ IP skip(□□ 10□). □□ □□□ AGENTS.md □ □□□ — env value □ □ / □□□ / deprecated alias □□ □□ □□ default □□ □□. fallback □□ □□□ □□ **AI □□ □□ 3□ follow-up** □□ □□□□ □□□ PR □□□ □□□□□□ — AI □□ □□□□ □□ □□□ □ □□□ □□□□□□ □□□□.



□□

- **□ □□ 800□ → 90□ □ (88.75% □□)** — □□ □□□ □□.
- **□□ □□□ 70% → 98%** — □□□ □□ □ □□ □□□□.
- □□ □□ SPOF □□(HA □□ □□ □□), □□□ □□□□ □□.
- □□ IP 가 legacy □□□ □□□□□ □□ **0**□.
- CROSSSLOT □□ **0**□ — hash tag {pool} □□ □.
- ASN □□ □□ + KR 가 + □□ □□ □□(failed outcome □□) □□.

JVM/Spring □□□ ([commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

- [ADR-SCR-010 Bulkhead □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-SCR-010-bulkhead.md) — Semaphore vs FixedThreadPool, □□ □□□□ □ □□
- [ADR-SCR-011 □□□□□ □□ □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-SCR-011-decorator-order.md) — Bulkhead → CB → Retry → TimeLimiter → RateLimiter (Retry □□ CB □□ CB □□ N□ □□□□□ □□)
- WebClient ExchangeFilterFunction □□ proxy resolver □□□ + @ConfigurationProperties + @Validated □ env □□

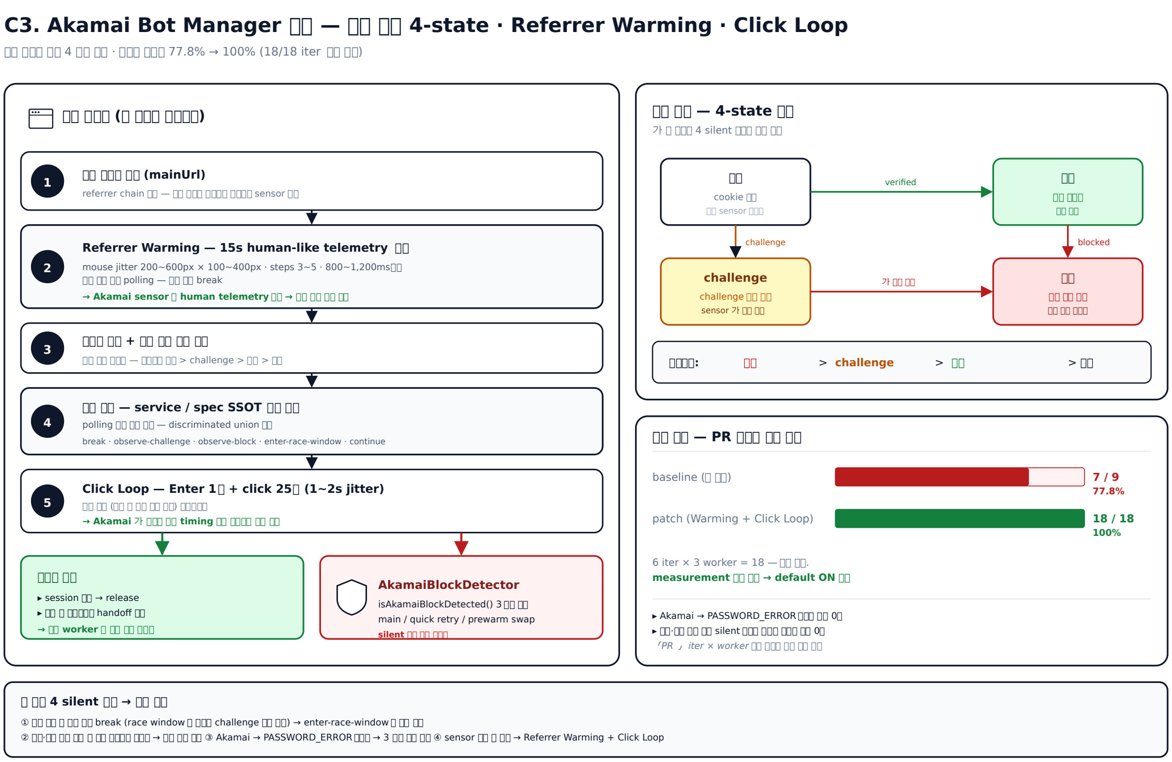
□□ 11. Akamai Bot Manager □□ — □□ □□ □□ □□ · Referrer Warming · Click Loop (77.8% → 100%) □ 2026.02

□□ — Akamai Bot Manager □ □□ □□ □□□□ □□ □□□□ □□□□□□ 99% □□. stealth fork □ fingerprint □ □□□□□ sensor □ □ □□□ □□ □ □□ □□□ □pending □□□ □□ sensor □□ □□ → 403.

□ □□□ 4 □ silent □□:

- □□ □□ □ □□ □□ break → race window □ □□□ challenge □□ □□
- □□ + 「 □ □□ □□□ □□□□□□
- Akamai □□ □□□ application-level 401/403 □□ □□□□ password retry □ □□ (Full Retry □□□)
- sensor □□ □ □□ break → sensor □ □□(1~2□) □□□ □□

□□ — □□ □□□□□ □ □□□ 4 □ □□□ □□: (a) □□ □□ **4-state** □□ □□□ □□□□ □□□ > challenge > □□ > □□□ + (b) □□ □□□ **service** □ **spec** □ **SSOT** □□ □□ □ □□(discriminated union □□ — break / observe-challenge / observe-block / enter-race-window / continue) + (c) **Referrer Warming** — □□ URL □□ referrer chain □□, 15 □□ human-like mouse jitter(200–600px × 100–400px × steps 3–5 × 800–1,200ms □□) □□ → sensor □ human telemetry □□ → □□ □□ □□ + (d) **Click Loop** — submit Enter 1 □ → page click 25 □(1~2□ jitter) □□□□ □□ □□ □□□□□ + (e) **3 □□ □□ □□** — Akamai □□ □□ helper □ main loginResult □□ + Quick Retry □□ + Prewarm Swap □□ □ □□□□ □□□□ □□ (□ □□ □□□□ □□□ set □□□ □□ silent □ □ □□□ □□□ □□□).



C3. Akamai □□ + □□ □□ 4-state □□

□□

- □□□ □□□ **77.8% → 100%** (Referrer Warming □□ □□, 18/18 iter, 6 iter × 3 worker, default ON □□).
- Akamai □□ → PASSWORD_ERROR □□□ □□ **0**□.
- □□·□□ □□ □□ silent □□□ □□□ □□□ □□ **0**□.
- PR □□□ iter × worker □ □□□ □□ □□□ □ □ □□ □□□□ □□□□ (7/9 → 18/18).

JVM/Spring □□□ ([commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

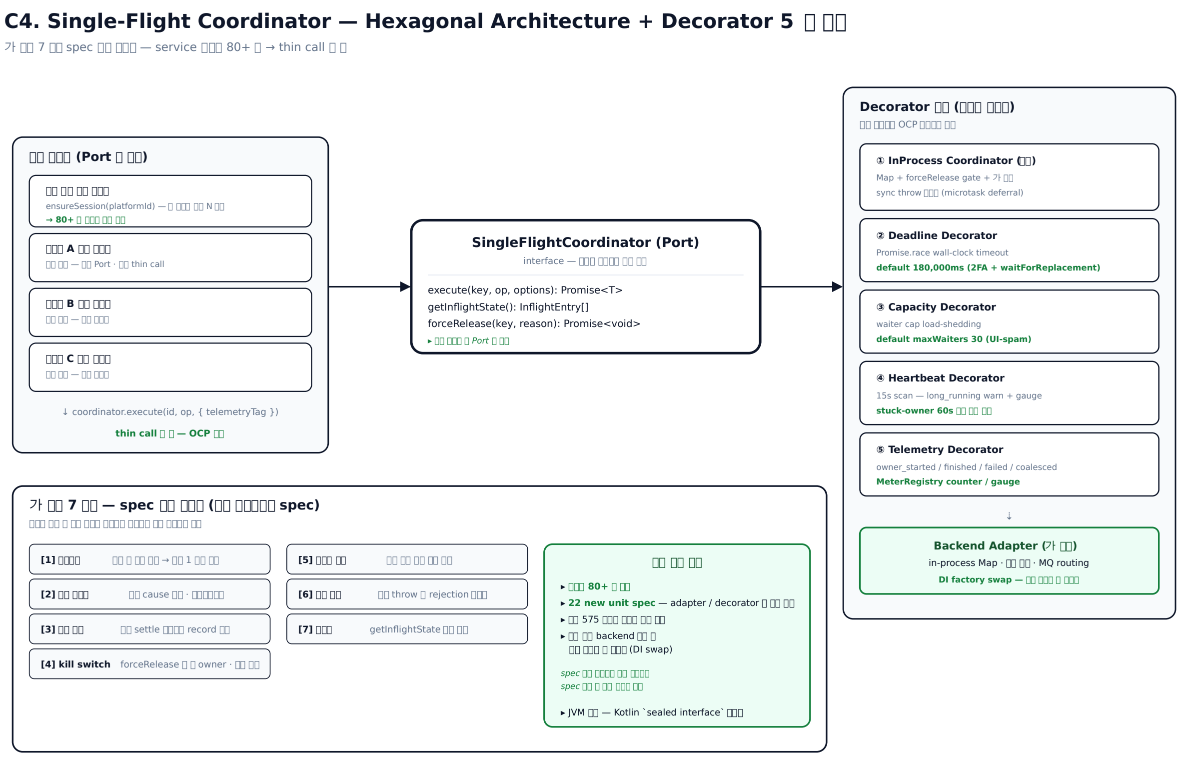
- Kotlin **sealed class** AbckState / PollingAction □□ discriminated union — when exhaustive □□□ □□□ □□ □□ □□
- WebClient interceptor □ □□ □□ □□ □□ □□ + [ADR-007 CB 70% □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-007-cb-threshold.md)
- ADR-SCR-010 □ □□ □□ □□ □□ □ □□ (Playwright / stealth fork)
- [EXP-WC-01 WebClient .block() EventLoop pinning □□□ 1/10 □□ □□](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/experiments/EXP-WC-01-webclient.md) — □□ □□□ □□

□□ 12. Single-Flight Coordinator □□□ — Hexagonal + Decorator 5□ □□, 7 □ □□□ □□ spec | 2026.01

□□ — □ □□□(platformId)□ □□ □□ □□ N□□ □□□ □ □□□ □□ □□□□□ □□□□ □□ □□□□ □□□ (single-flight · coalescing).

- □ □□□ □□ — □-□□□ Map 80+ □□ □ □□□□ □□□. □□ □□□ □□□□ □□ □□ □□ / □□ □□ □□□ □□ □ □□ □□□ □
- □□□□□ □□ **invariant** □ — □□ □□□ (□□ cause □□ · □□ □□□□ □□) / 가 □(forceRelease □ □ record finally cleanup □ □ record □□ □□) / □□ throw □□□ (sync throw □ Map □□ □ throw)
- □□ □□□ spec □ □□□□ □□ □□□ □□ □ □□

□□ — **Port-Adapter(Hexagonal Architecture, Alistair Cockburn) + Decorator + Strategy** □□ □□ OCP □□ + □□ □□□□□ **spec** □□ 7가 □□ □□□ spec □□□ □□□□□ describe □□□ 1:1 □□. Port □ execute / getInflightState / forceRelease □ □□, □□□□ in-process Map(forceRelease gate □□) + 4 □□□□□(Deadline = Promise.race wall-clock timeout / Capacity = waiter cap load-shedding / Heartbeat = 15s scan long_running warn + gauge / Telemetry = owner_started/finished/failed/coalesced metrics) □□. 가 □ □□this.inflight.get(key) === record □ □□ □□, sync throw □□□□ Promise.resolve().then(() => operation()) □ □□ □ hazard □□ □□. □□□ default □ deadline 180,000ms(2FA + waitForReplacement), maxWaiters 30(UI-spam load-shedding), long_running warn at 60s(stuck-owner early signal), forceRelease ops kill-switch.



C4. Single-Flight Coordinator — Hexagonal + Decorator 5□

□□ □□ 7□□ — spec □□□ □□□ (□□□□ □□ □□ □□□ □□□□ □□□□ □□ □□□□ □□)

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2	□□ □□□	□□ cause □□□□ □□ (□□ □□□□ □□)
3	□□ □□	□□ settle □□□□ record □□ □□
4	kill switch	forceRelease □□ □□ □□□ □ owner, □□ □□
5	□□□ □□	□□ □□ □□ □□ □□
6	□□ □□	□□ throw □ promise rejection □□ □□□
7	□□□	getInflightState 가 entry □□ □□

□□

- □□□ □□ **80+** □ □□ — coordination + telemetry 가 thin call coordinator.execute(id, op, { telemetryTag }) □ □□.
- **22 new unit specs** — adapter/decorator □ □□ □□, □□ □□□ □□□ □□ □□.
- □□ □□ backend 가 □ □□□ □□ □ □□□□ DI factory □ swap.
- □□□ □□□□□ spec □□ □□□□ □□□□ □□ □□□ □□ □□□□□□ □□ □□□ □□□.

JVM/Spring □□□ ([commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

TS □□□□□ 7가 □□ □□□ Kotlin Coroutines □ □□□ □□.

- [ADR-SCR-008 Single-Flight Coordinator](https://github.com/PreAgile/commerce-external-gateway-kt/blob/main/docs/adr/ADR-SCR-008-single-flight.md) — Mutex + Deferred □ in-process single-flight, 7-property □□ □□ spec □□ □□ □□
- Resilience4j Bulkhead.semaphore + TimeLimiter □ deadline / capacity
- SF-2 □□ — thundering herd baseline vs Coroutines coalescing □□ □□ **99.9% coalescing**
- MeterRegistry counter / gauge □ singleflight.* telemetry (owner_started / finished / coalesced)

□□□ D. AI □□□□□ · □□ □□□

□□ 13. □□ □□ □□ — 6□□ □□□ □□ + 가 □ □□ + □□·□□□ OR | □2026.01 ~ □□ □

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- □□□ 5□□□ □□□ □□□□ (sentiment □□□ — □ 30%가 5□ + □□ □□)
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□□ — □□ □□ + □□ □□ □□□ □□ □□ □□ □□ □□ if-else □□ □□ **6□□** 가□□ □□ □□□ □□□ □□ □□□ □□ □□ ML □□□ □□. □□ □□□ □□□□ **shrink (Bell)**□ □ □□□ □□ **(Polarity Amplification)**」 □ □ □□ □□ + **tanh ±35 sigmoidal squashing** 가 □□ □□□ □□ □□ □ □□□ □□□□.

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□□ □□□ (top word share)	top_word_count / total_tokens
Bigram / Trigram □□□	most_common_ngram / total_ngrams
□□ □□□□	-Σ p log2 p (□□ □□)
□□ □□□	unique_chars / length (40□ □□□□□)
zlib □□□	len(compressed) / len(text)
□□ □□ / □□□ 4□+ □□	□□□ □□ □□

□ □□ □□□ □□ □□ □□ 2.0 □□□ noise. □□ □□□□ Bell □□□ □□□ shrink + □□ □□ weight □□ □ tanh ±35 box □ squash. □□ □□□ □rating ≤ critical_rating□ □□ □□ OR □5□ + score in (min~max) + len ≥ 10□ scoring-based □□. □□□ **BIZ(□□□ `send_negative_alarms` + `critical_review_rating` □ □) □ User(`NegativeReviewAlarm.is_active`)** □ □□□ □□, □□□ □□□ □□□□□□ □□ □ □ □□ = □□ □□ □ □□□ □□□ □□ □□□. □□□□ □□□ SCORE_VERSION □□□ PK □□□□ □□ □□ review_id □ □□ □□ □□ → v2 □□ □ v1 □□ □□□□□ □ □□.

□□

- AI □□ □□ — existing_scores_map pre-fetch □ □□ □□ □□ □□ □□ + □□□ □□ □□□ □□ □□ ML □□ skip.
- **5□ □□ □□ □□□ □□** — □□ + □□□□ OR □□□□ □ 30% □ □□□ ≠ □□□ □□□ □□.
- □ □□ □□□ □□ BIZ 3□ + User 3□ = 6□ □□□ □□ □□.
- □□□ □□□ □□ □□ — □□ □□□(□□□·□□) + □□□□ □□ □□ Enum □□ □□.

JVM/Spring □□□ ([commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator))

- Spring Batch ItemProcessor □□ — 6□□ □□□ □□ + □□ □□□ + AI Score □□
- WebClient (□□□) □ □□ MLOps Score API □□, Bean Validation □□ □□ □□

- Kafka Consumer Group + DLQ + retry topic □□ □□ □□ □□ □□□□ □□□□□ ([ADR-MQ-008](https://github.com/PreAgile/commerce-batch-orchestrator/blob/main/docs/adr/ADR-MQ-008-consumer-idempotent.md))
- ApplicationEventPublisher + @TransactionalEventListener(phase=AFTER_COMMIT) □□ □□ □□□ □□□□ □ □□

□□ 14. SafeCron — □□ □□□□ cron □□ □□ □□ □□□□□ (14□ □□) □ 2025.09 ~ □□ □

□□ — □□ □□□□ □□□□ □□ @Cron □ □□ □□□□□□ □□ □□.

- □□ □□ □□□ □□□ N□ □□□□□□ N□ → □□ PG API rate-limit □□
- □□ □□□ cron N□ → □□□□ □□ □□□ N□
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□□

- □□□ **14□** (subscriptions, □□□, □□□ SaaS □□□, magic-link, mixpanel, log-cleanup □).
- □□ 5~10□□ □□□□□□□ □□ → □ **100□□** □□ + cron 가 □□□ □□ □ □□□□.
- □□□□ □ □ □□□ □ cron □□□□ — QA 가 □□□□ □□.□
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JVM/Spring □□□ ([commerce-batch-orchestrator](https://github.com/PreAgile/commerce-batch-orchestrator))

- Spring @Scheduled + **ShedLock** (Redis / JDBC backend) — □□ □□□□ cron □□ □ □□ □□ + ADR-005 Outbox polling-to-CDC □□ □□
- @ConditionalOnProperty □ □□□ toggle (QA □□)
- Spring AOP @Around □ Slack □□ / □□ / □□ □□ □□□

□□ 15. □□ □□ □□ □□□□□□ — □□ → AI → 4□ TOCTOU □□□ → API Key/JWT □□ □□ □□ □ 2026.03

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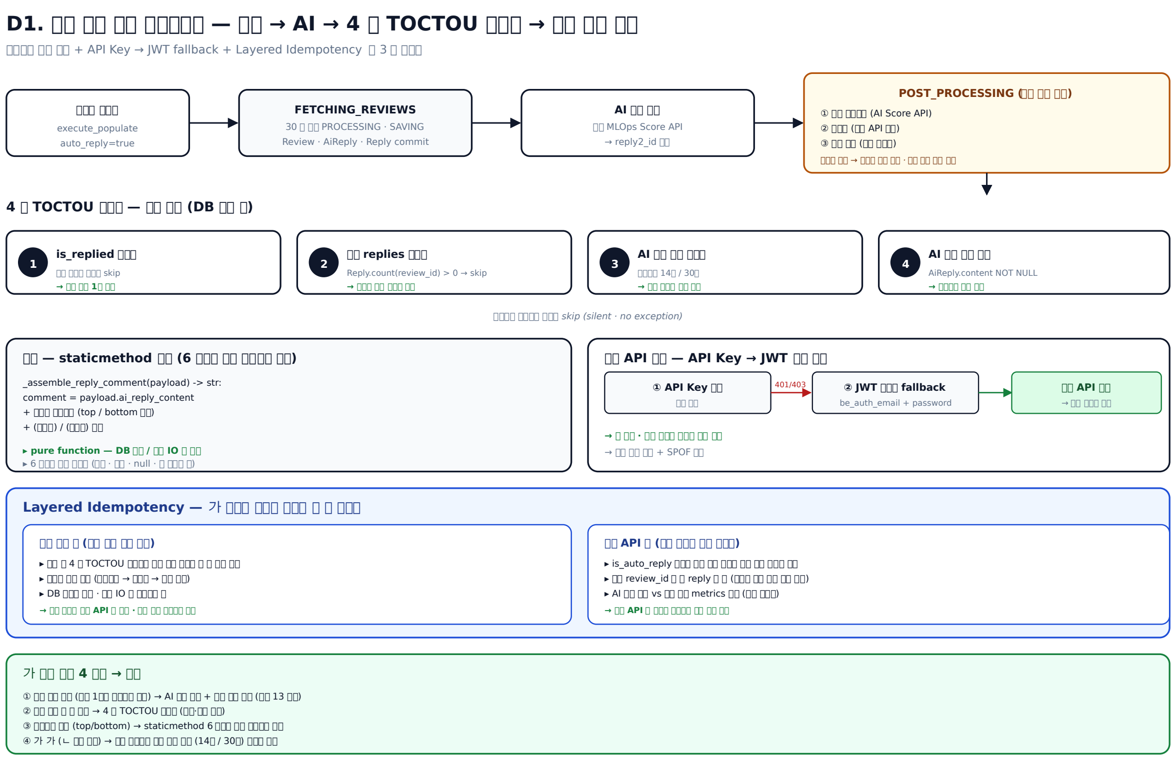

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Layered Idempotency □□ □□

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- API Key → JWT fallback □□ □ □□ □□□ □□□ □□ □□.
- AI □□ □□ vs □□ □□ □□ metrics 가 「 —AI □□ □□□ vs □□□ □□□ □ □□ □□.

JVM/Spring □□□ ([commerce-comment-platform-be](https://github.com/PreAgile/commerce-comment-platform-be) + [commerce-external-gateway-kt](https://github.com/PreAgile/commerce-external-gateway-kt))

- [ADR-BE-008 □□□□ □ □](https://github.com/PreAgile/commerce-comment-platform-be/blob/main/docs/adr/ADR-BE-008-transaction-boundary.md) — @Transactional(readOnly=true) □□ □□ + WebClient □□ □□□ □□□□ □ (EXP-09 9 □□□□ □ □□ □□)
- Spring Security OAuth2AuthorizedClientManager □ API Key → JWT fallback
- Resilience4j Retry + CircuitBreaker □□ + ADR-SCR-011 □□□□□ □□ □□

JVM □□□ □□□□ — Node.js □□ □□□ Kotlin/Spring □□ □□ + ADR □

5□ □□ Node.js □ □□□□ □□□ □□ □ · □□ □□□ · □□□□ □□ · □□ □□□ □□ □□□ Java/Kotlin/Spring □□□□ □□□ □□□□□ □□ □□□□, □□ □□□ ADR □ □□□ □□□□□□□□. □□□□ **ADR 49□, □□ 47□, □□ □□□ 17□** □□.

Node □□ □□ ↔ Kotlin/Spring □□□ □□□□

Node □□	**Kotlin/Spring □□□**
□□ □□ SETNX + Lua □□ □	Redisson watchdog + Pub/Sub 4□ □□
Idempotency Key + □□□□ + □□ + reconciliation	ADR-006 □□ □□□ 4□□ + EXP-09b 9 □□□□
□□ □ □□□□□ FIFO + lease TTL	Coroutines + supervisorScope + Single-Flight 5 invariants
Custom Error Strategy 5□ + 9□ □□□□	Resilience4j 5□ + 9□ □□ □□□
□□□□ □□□□ PID Registry (OOM □□)	JVM Heap / GC □□ G1GC Region + Humongous
□□□ □ classic queue (□□ 5가 □□ □□)	Kafka □□□□□□ + Outbox Relay

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- □□·□□ □□□ □□□ (Java / Spring Boot / JPA) — □□ □□□ 4□□ □□, □□ □ **4**□ □□(□□□ · □□□ · GET_LOCK · Redisson), DB InnoDB RR vs ANSI RR □ □□ □□, □□□□ □□ □□ 9 □□□□ □□□□, No-offset Pagination.
- □□ □□□□□□□ □□□ (Spring Batch + Kafka) — □□□ □□ 1□□ □□□ 5가 □□(replay □□, prefetch HoL, x-overflow, publisher confirm □□, DLQ □□ □□)□ □□□□ Kafka □□□ □□□□□ ADR □□. Outbox Relay □ polling vs CDC □□ □□, Spring Batch Reader 4□ □□□□.
- □□ □□□□□ □□□ (Kotlin / Coroutines / Resilience4j) — □□ □□□ Kotlin □□ □□□. **Single-Flight** □ □□ □□ □□□(promise sharing, sync throw normalization, force release, deadline, capacity), 9 □ □□ □□□□ □□□, Resilience4j 5 □□(CircuitBreaker, Retry, Bulkhead, RateLimiter, TimeLimiter) □□.

[illegible] — [illegible](I-BRICKS) | | [illegible] 2021.05 ~ 2024.11 [illegible]

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]·[illegible] [illegible], [illegible] [illegible], [illegible] [illegible] [illegible].

- [illegible] [illegible] [illegible] [illegible]·[illegible] — Elasticsearch + Logstash [illegible] RDB → ES [illegible] [illegible], [illegible] [illegible] [illegible]·[illegible] [illegible]가 [illegible] [illegible] [illegible]·[illegible] [illegible] [illegible] [illegible] [illegible].
- **EBS [illegible] [illegible] [illegible] [illegible]** — [illegible] [illegible] [illegible] [illegible] Kafka + Apache Nifi + Elasticsearch [illegible] [illegible]. [illegible] [illegible] [illegible] **50%** [illegible], [illegible] [illegible] 2가 [illegible] [illegible].
- [illegible] [illegible] [illegible] — React / Redux / SCSS [illegible] [illegible] UI, KWCAG 2.1 [illegible] [illegible] [illegible].

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- □□□ □□ □□ □□□ □□ — □□□ DB □□ □□ □ + SELECT FOR UPDATE □□ □□ + □□ □□ Pub/Sub □□ □□ + □□ □□□ □□ □□□□ **TPS 100 → 1,000+, □□ □□ 5□ → 10□ □□, □□ □□□ 1%** □□ □□.

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- □□□ □ DLX □□ application-level retry — setTimeout + republishMessage □ hot loop □□, 30s / 5m / 30m □□□ backoff. Phase 2 □□ DLX □□□ □□ □□□□□□ □□ □□.
- □□ □□ IntegrityError-aware upsert + Lock-timeout □□ □□□ — Duplicate □ break, Lock timeout □ 1s / 2s / 3s □□ backoff, □□ □□ raw SQL ON DUPLICATE KEY UPDATE □□.

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- □□ API quarantine — DB-backed state machine(OK / QUARANTINED) + 12h probe □□ □□ + DB↔□□ □□ mismatch □□.
- □□ □□□□ □□ □□ + □□ □□ □□□□ — □□ □□(□□ □□□ □□□ ++) / □□ API □□ 2□□, □□ 3□ □□ □□□□ + □□ □ □□□ □□ □□ □□.
- □□□ □□ □□ □□□ — □□□□ 18□ + 11□+ □□□ BasePlatformException → □□□□ □□□ → classifier helper □ 3 layer □. instanceof + name + code 3□ □□□ □□□ □□ □□.
- Idempotent Reply Pipeline — Lease + Completion Cache □□ + 4□ ErrorClassifier(BLOCKED_SUSPECTED / TRANSIENT / PERMANENT / SYSTEM) + ACK/NACK □□□□ □□□. □□□ PERMANENT(unknown □□ □□ retry □□).
- Adaptive Traffic Controller + Launch Budget — Lua Token Bucket + Circuit Breaker(SOFT_OPEN / HALF_OPEN) + jittered watchdog □□ Thundering Herd □□.

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- □□□□ □□□□ □ — Launch Semaphore + Prewarm Pool + Quarantine + 5-phase Zombie Cleanup + Jittered Watchdog 5□ □□ □□. 40 concurrent □□ **97.2%** □□□.
- OOM □□ chunk □□ — □□ □□ □□ cron □ find □□□ □□□ CHUNK_SIZE=2000 while □□ + lookup map 1□ □□□□ □□.
- □□□ □□□□ □□□ — □□□ □□□□ 30□ □□ + Semaphore 30 + □□ DB □□ + □□□ gc + 3□□ □□ □□□□□ □□ □ □□□. □□□ □□□□ □□ gc □□□ □□□.

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- □□ □□□□ □□□□□□□ — □□ □□□□□□□ □□ □□ Hash + ZSet(□□□ / □□□ □□□) + TTL □□□ □□ (RUNNING 6h / COMPLETED 1h / FAILED 2h)□ □□. □□ □□□ □ orphaned job □□ □□□ (Semaphore 10 □□□ □□).
- 24□□ □□ □□□ + Config □ RUNNING 1□ — 24□□ □ □□□ DB □ □□ □□ ±5□ □□□ dedupe □ □□ 0□, Config □ RUNNING 1□ □□ + □□ □□ Semaphore(200).

- [illegible] backfill — [illegible] + [illegible] fetch in-memory set [illegible], [illegible].

[illegible] · [illegible]

- [illegible] + [illegible] SaaS [illegible] — [illegible] → [illegible], [illegible] + 5[illegible] retry queue [illegible].
- APM Span Tagging — tagDatadogSpan [illegible]([illegible], [illegible] amount mismatch)[illegible].

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□□	TypeScript, JavaScript, **Kotlin, Java**, Python
□□□□□	NestJS, Node.js, **Spring Boot, Spring Batch**, FastAPI
ORM	TypeORM, Python ORM, **JPA / Hibernate**, APScheduler
□□□□□□	□□□ DB (Cluster, Replica) — MySQL □□ / PostgreSQL / Oracle
□□ · □	□□ □□ (Cluster □□), Lua eval, Token Bucket, **Redisson**
□□□	□□□ □ (Quorum Queue, DLX, classic queue □□ □□), **Kafka**
□□	Elasticsearch, Logstash, Apache Nifi
□□ □□□ □□	**Resilience4j**, Custom Retry / Timeout, Single-Flight (Port-Adapter), Circuit Breaker (SOFT_OPEN / HALF_OPEN), Token Bucket
□□□□ □□□□	stealth fork □□ □□□□ □, □□ □□ 5□
□ □□ □□	Akamai · Cloudflare □□ — □□ □□ 4-state □□, □□ □□ polling, Referrer Warming, Click Loop jitter
□□ · □□	Prometheus, Grafana, **APM (span tagging)**, nGrinder, JFR
□□□	Jest, **Testcontainers, JUnit 5, Kotest**, Pytest, "□□ spec □□ □□□" □□□
□□□	Docker, Docker Compose × 8 color, □□ □□□ □□, □ □□□ □□□□□□□□, □□ Cloud, Jenkins, GitHub Actions
□□□□□□	Flyway, Liquibase □□
□□	Slack Webhook (Block Kit), □□□ OAuth2 + □□□ □□ □□□
AI / ML	□□ MLOps Score API + 6□□ □□□ □□ (Shannon entropy, zlib □□□, n-gram, emoji run) + 가 □ □□ (Bell + Polarity Amplification + tanh box)
□□ □□□ □□	Redlock(Martin Kleppmann) ABA □□ fencing token, Hexagonal Architecture(Port-Adapter), CQRS, Saga, Outbox, CAS, Bulkhead, Circuit Breaker, Single-Flight

[illegible]

F-Lab Java Backend Mentoring | 2024.01 ~ 2024.07

Meta [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible], [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible], CQRS, [illegible] [illegible] [illegible] [illegible].

[illegible] [illegible] [illegible]